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Literature Review

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Writing Interventions for Students With Intellectual and Developmental Disabilities: A Meta-Analysis

Derek B. Rodgers, PhD[1], and Susan J. Loveall, PhD[1]

Abstract

Students with intellectual and developmental disabilities (IDD) are likely to experience writing challenges that can negatively impact their overall academic, vocational, and social success. The goals of this study were to determine what writing interventions have been used with students with IDD, how their writing skills have been measured, the effects of those interventions, and study-level moderators via meta-analytic procedures. We identified 96 writing-related outcomes reported in 52 studies that included 424 students with IDD. A majority of studies used strategy instruction/self-regulated strategy development (SRSD), direct instruction, or response prompting to instruct writing skills, and the most common outcome measures were related to writing output, conventions, quality rubrics, elements, and spelling. Results indicated that writing interventions led to significant improvements in students' writing skills. These effects were moderated by intervention type and study quality.

Keywords

intellectual disability, exceptionalities, intervention, written language, meta-analysis, research methodology, autism

Written expression is an essential part of academic learning for all students at all grade levels, and strong writing skills are associated with academic, professional, and social success. For example, teachers use student writing to assess learning across different topics (Graham & Hebert, 2011), and writing skills are positively correlated with vocational outcomes (National Commission on Writing in America's Schools and Colleges, 2004). Furthermore, the ubiquity of social media means that an individual's ability to write clearly is more critical than ever (Penner-Williams et al., 2009). Thus, writing is an avenue toward independence and social engagement, and may be especially important for individuals with intellectual and developmental disabilities (IDD), who often experience difficulties finding postsecondary education opportunities, employment in the community, and independent living options after high school (e.g., Shogren & Plotner, 2012).

Students with IDD are a diverse group of learners and include those with an intellectual disability (ID; i.e., an IQ below 70/75 and significant limitations in adaptive functioning), autism spectrum disorder (ASD) with or without ID, and other developmental disabilities (e.g., Down syndrome). These students often experience difficulties developing writing skills (Woods-Groves et al., 2018) and also often have working memory and self-regulatory skill deficits that can impact their acquisition and transfer of academic skills (Asaro-Saddler & Saddler, 2010; Gargiulo & Bouck, 2018).

Writing Skill Development

The Levels of Language framework (Berninger et al., 2009), a developmental model of writing, is a useful lens through which to examine the hierarchical nature of writing skills. The framework presents written language across the letter (handwriting), word (spelling), sentence (grammar and mechanics within sentences), and discourse levels (larger compositions; Berninger et al., 2009). Students typically master letter- and word-level writing skills first, then progress to using sentence and discourse-level writing skills.

Depending on their present skill set, students with IDD may need instruction at any one of the levels of language. For example, they may have trouble with spelling, syntax, grammar, and/or sentence construction (Rodgers & Datchuk, 2020; Monroe & Troia, 2006). Similarly, students with IDD

[1]University of Nebraska-Lincoln, USA

Corresponding Author:
Derek B. Rodgers, Department of Special Education and Communication Disorders, University of Nebraska-Lincoln, Lincoln, NE 68502, USA.
Email: derek-rodgers@uiowa.edu

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often show challenges with planning, generating, organizing, and reviewing lengthier compositions (Asaro-Saddler & Bak, 2012, 2014; Woods-Groves et al., 2018).

Prior Writing Intervention Meta-Analyses for High-Incidence Disabilities

There is a rich literature examining writing interventions for students with high-incidence disabilities, including many meta-analyses (e.g., Kaldenberg et al., 2016; Datchuk et al., 2020; Gillespie & Graham, 2014; Graham et al., 2016, 2017; Graham & Harris, 2003). Meta-analysis is a rigorous method of synthesizing weighted effect sizes (Borenstein et al., 2009). This allows researchers to evaluate the impact of moderators (e.g., length and quality of intervention) on the overall results. In high-incidence disabilities, these meta-analyses report that writing intervention effectiveness varies depending on the type of intervention used (Gillespie & Graham, 2014; Graham et al., 2016), writing task(s) completed (Datchuk et al., 2020), and study quality (Gillespie & Graham, 2014).

Unfortunately, students with IDD are underrepresented in broader disability research (Bouck, 2017). A comprehensive and quantitative aggregation of the writing literature for students with IDD would help professionals identify what interventions have been used at what levels of language and the degree to which they were effective. This knowledge would help build a concise list of evidence-based writing practices for these overlooked learners.

Prior Reviews on Writing Interventions for Students With IDD

Although no meta-analyses have been conducted, several systematic reviews have identified direct instruction, response prompting, and strategy instruction as effective instructional techniques for teaching writing to students with IDD (e.g., Asaro-Saddler, 2016; Joseph & Konrad, 2009; Pennington et al., 2011; Pennington & Delano, 2012; Rogers & Graham, 2008). Self-regulated strategy development (SRSD) has been the most well-researched and appears to be the most effective method of promoting independence across numerous writing skills (Joseph & Konrad, 2009; Rogers & Graham, 2008). SRSD is a specific six-stage instructional approach that typically uses a criterion-based performance, requiring students to master a specific—usually mnemonic—strategy (e.g., Pick my ideas, Organize my notes, and Write and say more; POW) before ending intervention (Graham & Harris, 2003).

Although informative, these prior systematic reviews are limited in two key ways. First, none were exhaustive reviews of the literature. Joseph and Konrad (2009) focused on students with IDD, but they identified only nine studies. Other reviews examined either specific subgroups of students with IDD (e.g., ASD without accompanying ID; Asaro-Saddler, 2016) or focused more broadly on all students with disabilities (e.g., Rogers & Graham, 2008). In addition, previous reviews all described different sets of primary studies, and there was almost no overlap in the studies reported across those reviews. This suggests that the literature has not yet been comprehensively aggregated in a single analysis.

Second, these reviews did not quantitatively aggregate the effects of the included writing interventions. Perhaps one reason for this is that single-case designs (SCDs) are commonly used for low-incidence populations such as IDD, which historically presented obstacles for quantifying intervention effects. Namely, most SCD effect sizes focus on non-overlap metrics (see Parker et al., 2011 for a description and Joseph & Konrad, 2009; Rogers & Graham, 2008 for examples of use in prior reviews), but these are partially insensitive to magnitude of change and often do not include a variance estimate. The lack of variance prevents the weighting of effects, limiting their utility for proper meta-analysis. Fortunately, a new between-case standardized mean difference (BC-SMD) for SCD studies yields an SMD on the same scale as Hedges's *g* (Hedges et al., 2012; Pustejovsky et al., 2014; Valentine et al., 2016). The metric has been used in recent meta-analyses (e.g., Ennis & Losinski, 2019) and could be useful for quantifying the effects of writing interventions for students with IDD.

Present Study

The purpose of this study was to identify and evaluate the effectiveness of the writing interventions that have been used with students with IDD. Specifically, we asked the following research questions:

Research Question 1: What are the interventions that have been used with students with IDD and how were writing skills measured?
Research Question 2: What are the effects of those interventions?
Research Question 3: How do study characteristics moderate those effects?

Method

Inclusion Criteria

Studies were included in the meta-analysis if they (a) described an intervention designed to improve a writing skill, (b) included participants with IDD who were enrolled in a full-time academic program (kindergarten through postsecondary), and (c) provided adequate data to calculate

effect sizes. With respect to the first criterion, we were interested only in interventions that included writing instruction and excluded studies that compared technologies without accompanying instruction. Both SCD and group design studies (e.g., randomized control trials or quasi-experimental studies) were included. However, SCD studies needed to include at least three participants with IDD because the across-individual variance that is essential to the BC-SMD is unstable when studies include fewer than three individuals (Valentine et al., 2016). Therefore, studies with fewer than three participants with IDD were excluded (e.g., Datchuk & Kubina, 2017). In addition, SCD studies could not use a traditional multiple baseline (or probe) across settings or behaviors design wherein a participant is observed in several unique settings with several unique measures. In addition to the three-participant requirement previously stated, the BC-SMD requires that participants must be observed with identical measures to be pooled together. Multiple baselines across settings or behaviors often include participants in multiple settings with outcomes unique to each participant, or they can involve measuring students' skills (behaviors) across multiple prompt types that are not equivalent across all participants in the study (see Purrazzella & Mechling, 2013). These individualized designs, although methodologically sound, are incompatible with calculating the BC-SMD and were therefore excluded from this analysis (Valentine et al., 2016). Group studies needed to include a control group to facilitate the calculation of the Hedges's *g* congruent with the BC-SMD. Because no other meta-analysis has been conducted for writing interventions for students with IDD, we did not place any limits on the publication date of papers to be included in the analysis, nor did we place any restrictions on the type of targeted writing skills. We included dissertations but not Master's theses.

Literature Search

The literature search was conducted in four phases. First, we ran an electronic search of four databases: ERIC, PsycInfo, Dissertations & Theses (ProQuest), and Academic Search Premier. To capture a broader literature base, we used a combination of disability and writing *subject* terms specific to each database instead of *search* terms. For example, ERIC indicates that users should adopt "mild intellectual disability" as a subject term to identify any articles with the key words "mental retardation," "intellectual disability," and so on. The disability subject terms used were *intellectual and developmental disability, developmental disability, mild/moderate/severe intellectual disability, cognitive impairment, cognition disorder, autism spectrum disorder, pervasive developmental disorders*, and *Down syndrome*. Writing-related subject terms were *handwriting, spelling, grammar, sentences, sentence construction, paragraph composition, essay writing, writing instruction, writing research, writing achievement, writing strategies, writing achievement*, and *written communication*. All identified study abstracts (n = 2,337) were screened by the first author and a secondary coder. Two coders were trained on a subset of abstracts prior to coding; agreement between the primary and secondary coders was 93%. All discrepancies were reconciled, and 142 studies were retrieved for further review. Of those, 44 articles were included. The other 98 studies were excluded because they did not include students with IDD (n = 14), did not include a writing intervention (n = 11) or outcome (n = 9), included insufficient data to calculate an effect size (n = 37), were not published in English (n = 6), insufficiently reported participants and intervention (n = 3), or were not a dissertation or published paper (n = 3). In addition, 13 SCD studies were excluded because they either contained too few participants (e.g., Datchuk & Kubina, 2017) or used a design incompatible with the calculation of the BC-SMD (e.g., Purrazzella & Mechling, 2013). Two group studies were excluded because they did not include a control group.

Second, we conducted a hand search of the following journals: *Education and Training in Autism and Developmental Disabilities*, *Exceptional Children*, *Journal of Developmental and Physical Disabilities*, *Reading and Writing Quarterly*, *Remedial and Special Education*, *Research in Developmental Disabilities*, *The Journal of Applied Behavior Analysis*, and *The Journal of Special Education*. We identified two additional articles.

Third, we conducted an ancestral and forward search of the already included papers. The ancestral search resulted in one additional paper. The forward search, conducted using Google Scholar, identified four additional papers.

Fourth, we conducted an ancestral review of several prior reviews (Asaro-Saddler, 2014; Joseph & Konrad, 2009; Pennington & Delano, 2012; Rogers & Graham, 2008) that were identified during the literature search. No additional studies were identified. The search and coding procedures concluded in July 2021.

The lead author also included one self-authored paper relevant to the analysis that was accepted for publication but not yet published. Therefore, we identified 52 studies described in 51 papers (40 SCD studies, 12 group studies) that yielded 96 writing-related outcomes.

Study Coding

Studies were coded for the following participant demographics: sex, age, race/ethnicity, grade level, disability status, and IQ score. We also coded for study design, setting, level of language, composition genre, SRSD status (i.e., 0 = not an SRSD study, 1 = SRSD study), independent variable, dependent measure(s), and number of intervention

sessions. We used number of intervention sessions as a proxy for intervention length because too few studies reported intervention length (i.e., only 33 of 52), but all studies reported the number of intervention sessions. For SCD studies, we averaged the number of intervention sessions across participants within each study. The first author coded all studies, and all studies were double coded by one of two secondary coders. Prior to coding, the two secondary coders were trained on a subset of articles until reaching 95% agreement. Overall agreement was 91%. Discrepancies were discussed and reconciled prior to analysis.

Because we identified more than two dozen unique writing interventions, we re-coded independent variables into one of eight categories partially adapted from Gillespie and Graham (2014): direct instruction, modeling alone, peer mentoring, pre-writing, systematic reinforcer delivery, response prompting, self-monitoring, and strategy instruction. Modeling was a component of several categories (e.g., direct instruction, strategy instruction, response prompting), but the modeling alone category represented studies that used modeling as the primary instructional technique and did not include components necessary to be categorized as one of the other options (e.g., errorless learning procedures, strategy instruction). Similarly, we identified 47 unique writing outcomes. We re-coded all outcomes as one of 10 categories, nine of which were identified in prior writing intervention meta-analyses (e.g., Graham et al., 2017). These nine categories were conventions, elements, handwriting, ideation, quality, spelling, organization, output, and voice. The tenth category, called procedures, described how well students adhered to strategy steps when writing.

Study Quality

We categorized studies using the What Works Clearinghouse (WWC, 2020) standards. The standards are rigorous, focus on a broad set of important factors (e.g., sampling, baseline equivalence), and yield identical quality categories for group and SCD studies: Meets Standards without Reservations, Meets Standards with Reservations, Does Not Meet Standards. See Table 1 for an overview of studies, including their quality designation.

Single-Case Design Studies

We calculated the BC-SMD, an SCD equivalent to Hedges's *g* (Pustejovsky et al., 2014), for each SCD writing outcome. The BC-SMD requires the specific values for each data point included in the analysis, so we used WebPlotDigitizer (Rohatgi, 2014) to extract those specific values from SCD graphs. WebPlotDigitizer has been used in previous SCD meta-analyses (Maggin et al., 2017) and yields higher reliability than similar programs (Moeyaert et al., 2017).

Group Design Studies

We used Hedges's *g* (Hedges, 1981) for each group study effect size. Two studies (Gladis, 1970; Haviland, 1982) did not report means and standard deviations but did report *F* and *t* statistics, respectively. We converted these to SMDs as described by Lipsey and Wilson (2001). Those SMDs were then adjusted using Hedges's *g*.

Data Analysis

All data analysis was conducted using R 3.6.0 (R Core Team, 2019), primarily with the *metafor* package (Viechtbauer, 2010). We used a random-effects model given the variability in study characteristics. Random effects models presume that the observed heterogeneity stems from both within-study and between-study variance (Borenstein et al., 2009). We aggregated SCD and group design effect sizes consistent with recent meta-analyses and prior research (e.g., Ennis & Losinski, 2019; Pustejovsky et al., 2014). Twenty-four studies contributed multiple effect sizes to the analysis, so we used robust variance estimation (RVE; Hedges et al., 2010) via the *clubSandwich* package (Pustejovsky, 2020) to correct for dependency. All aggregate effect sizes are reported alongside the prediction interval (PI), which provides a range within which the true effects are distributed related to the summary effect (see Borenstein et al., 2009).

Sensitivity Analyses

We explored two potential sources of bias prior to calculating the summary effect: (a) differences in average effect sizes between SCD and group design studies, and (b) differences in average effect sizes between SCD studies that did and did not collect intervention phase data. For the first, we hypothesized that SCD design studies might yield systematically larger effect sizes because there are fewer participants who often receive individualized instruction. For the second, we identified that some studies did not collect intervention phase data (i.e., they only reported baseline and post-intervention/maintenance data). We hypothesized that omitted data points could introduce bias into the aggregate effects as prior research has shown that fewer SCD data points can result in larger levels of bias and overestimates of true variance (Pustejovsky et al., 2014). We therefore compared SCD studies that did and did not collect intervention phase data.

Additional Subgroup and Meta-Regression Analyses

As in prior studies (e.g., Gillespie & Graham, 2014), we were interested in exploring descriptive effects for different independent and dependent variables. Therefore, we examined the summary effects for type of intervention, type of

Table 1. Characteristics of Studies Included in the Meta-Analysis.

	Participant characteristics				Intervention characteristics			
Study	Grade	Sex	Race	Diagnoses	IV	DV	SMD (*SE*)	Quality
Allen-Bronaugh (2013)	MS	1 F 5 M	6 W	6 ASD	Strategy Instruction (SRSD)	Persuasive Essay Parts Essay Quality Score TWW # of Sentences # of Sentences # of Transitions	2.80 (0.60) 2.65 (0.48) 1.54 (0.39) 2.55 (0.52) 2.45 (0.53) 1.81 (0.38)	DNM
Almumen (2017)	MS	6 M	2 B 4 W	6 ASD	Strategy Instruction	# of Words per Essay # of Quality Indicators Overall Essay Quality	1.39 (0.37) 8.36 (2.50) 5.92 (1.67)	MWR
Asaro-Saddler (2014)	ES	3 M	3 W	3 ASD	Strategy Instruction (SRSD)	# of Story Elements Holistic Story Quality TWW	7.03 (1.31) 4.43 (0.94) 1.95 (0.65)	M
Asaro-Saddler & Bak (2012)	ES	1 F 2 M	3 W	3 ASD	Strategy Instruction (SRSD)	Holistic Quality TWW	4.03 (0.99) 0.74 (0.53)	MWR
Asaro-Saddler & Bak (2014)	ES	1 F 5 M	NR	4 ASD, 2 ASD/ID	Strategy Instruction (SRSD)	# of Essay Elements Holistic Essay Quality TWW	2.66 (0.54) 1.93 (0.44) 1.76 (0.40)	DNM
Asaro-Saddler & Saddler (2010)	ES	3 M	3 W	3 ASD	Strategy Instruction (SRSD)	# of Story Elements Holistic Story Quality TWW	5.03 (1.51) 2.10 (0.75) 0.82 (0.71)	DNM
Bailey et al. (2017)	MS	2 F 18 M	NR	20 ASD	Direct Instruction	Spelling Accuracy SSS—Elements Spelling SSS—Words Spelling	−0.03 (0.43) 0.40 (0.43) 0.47 (0.44)	MWR
Bishop et al. (2015)	MS	3 M	1 Multiracial 2 W	3 ASD	Pre-Writing	Correct Writing Sequences Total Words Written Analytic Rubric	0.78 (0.47) 1.08 (0.45) 3.99 (1.28)	MWR
Bonanno (1974)	MS	1 F 3 M	NR	4 ID	SRD	% of Correct Responses	0.94 (0.41)	DNM
Brady (2018)	HS	1 F 2 M	2 B 1 Multiracial	1 ASD/ID, 2 ID	Strategy Instruction	TWW # of Sentences # of Transition Words Holistic Quality Rubric	1.92 (0.55) 4.27 (0.70) 4.90 (1.55) 1.92 (0.55)	MWR
Delano (2007)	HS	3 M	NR	3 ASD	Strategy Instruction (SRSD)	TWW # of Essay Elements	0.19 (0.27) 1.87 (0.83)	DNM
Evmenova et al. (2016)	MS	4 M	1 A 1 H 2 W	4 ASD	Strategy Instruction	# of Essay Parts Holistic Rubric	4.32 (0.92) 3.81 (0.75)	M
Gladis (1970)	MS	20 F 36 M	NR	56 ID	Direct Instruction	Spelling Accuracy	0.27 (0.41)	DNM
Guzel-Ozmen (2006)	MS	4 M	4 W	4 ID	Strategy Instruction	Analytic Rubric	4.86 (1.31)	MWR
Harris et al. (2017)	ES	1 F 2 M	1 B 2 W	3 ASD	Modeling Alone	Legibility Score	1.48 (1.66)	M
Haviland (1982)	HS	12 F 10 M	NR	22 ID	Response Prompting	Mean # of Words per T-Unit	0.41 (0.37)	DNM
Jackson et al. (2018)	PS	1 F 2 M	3 W	3 ASD	Strategy Instruction (SRSD)	Quality Rubric	0.03 (0.20)	DNM
Joseph (2002)	ES	2 F 1 M	2 B 1 W	3 ID	Direct Instruction	# of Correctly Spelled Words	2.50 (0.52)	M
Kaldenberg (2015)	PS	11 F 10 M	1 A 1 B 19 W	2 ASD/ID, 9 ID, 2 DD, 8 Other	Strategy Instruction	Simple Sentence Quiz Sentence Writing Task Analytic Rubric CWS	0.79 (0.10) 0.21 (0.10) −0.46 (0.09) 0.00 (0.09)	DNM
Konrad et al. (2017)	HS	5 M	1 B 4 W	5 ID	Strategy Instruction (SRSD)	Expository Paragraph Quality Score IEP Paragraph Quality Score IEP Paragraph Content Score	0.63 (0.29) 3.24 (0.89) 3.21 (0.70)	MWR

(continued)

Table 1. (continued)

	Participant characteristics				Intervention characteristics			
Study	Grade	Sex	Race	Diagnoses	IV	DV	SMD (*SE*)	Quality
Liberty & Fitzpatrick (1994)	HS	4 F 2 M	NR	6 ID	Writing Process	# of Complex Ties	1.09 (0.25)	DNM
Lim & Lee (2019)	MS	3 M	3 A	3 ID	Writing Process	Writing Performance Score Content Score # of Letters	0.56 (0.37) 0.26 (0.34) 0.29 (0.35)	MWR
McLaughlin & Walsh (1996)	NR	NR	NR	6 ID	Modeling Alone	% Correct Letters	5.02 (1.21)	DNM
Mims et al. (2017)	HS	1 F 2 M	3 W	3 ID	Response Prompting	% Unprompted Steps Completed	0.97 (0.35)	MWR
Panos (2019)	ES	3 M	1 B 2 W	3 ASD	Direct Instruction	Correct Letter Points Error Letter Points	0.66 (0.45) 0.33 (0.27)	MWR
Park et al. (2017)	MS	1 F 2 M	NR	3 ID	Direct Instruction	% of Writing Quality Points	1.04 (0.56)	MWR
Pennington et al. (2011)	ES	3 M	3 W	3 ASD	Response Prompting	% Story Steps Completed	0.07 (0.24)	M
Pennington, Delano, & Scott (2014)	HS	3 M	NR	1 ASD/ID, 2 DD	Response Prompting	# of Cover Letter Components	1.31 (0.53)	M
Pennington & Koehler (2017)	MS	3 M	3 W	1 ASD/ID, 2 ID	Modeling Alone	# of Story Elements	1.18 (0.48)	M
Pennington et al. (2020)	ES	3 M	3 W	3 ID	Response Prompting	% Correct Sentences	1.05 (0.48)	M
Pennington & Rockhold (2018)	ES	4 M	1 B 3 W	4 ASD	Response Prompting	% Correct Sentences	1.36 (0.35)	M
Pennington, Saadatzi, et al. (2014)	HS	2 F 1 M	3 W	1 ASD/ID, 2 ID	Response Prompting	# of Text Message Components	0.63 (0.38)	M
Pennington et al. (2021)	NR	3 F 4 M	3 B 4 W	3 ASD/ID, 4 ID	Response Prompting	% Correct Sentences	0.98 (0.27)	MWR
Preast (2009)	NR	2 F 4 M	4 B 2 W	6 ID	Direct Instruction	Correctly Spelled Words	1.16 (0.34)	DNM
Rago (2013)	MS	1 F 4 M	1 A 3 H 1 W	4 ASD 1 ASD/ID	Strategy Instruction	% Correct Sentences	1.13 (0.30)	MWR
Rodgers & Datchuk (2020)	HS	2 F 2 M	2 B 2 W	2 ASD/ID, 2 ID	Direct Instruction	CWS IWS Paragraph Rubric	0.62 (0.28) 0.78 (0.32) 1.24 (0.40)	MWR
Rodgers et al. (2020)	PS	1 F 3 M	1 A 3 W	2 ID, 2 DD	Direct Instruction	CWS IWS	0.80 (0.27) −0.06 (0.18)	MWR
Rodgers et al. (2022)	HS	2 F 1 M	3 W	2 ASD, 1 ID	Direct Instruction	CWS IWS Paragraph Rubric	0.40 (0.30) 0.47 (0.28) 1.63 (0.50)	M
Rousseau et al. (1994)	MS	1 F 2 M	NR	3 ASD/ID	Direct Instruction	# of Adjectives per T-Unit	0.89 (0.33)	M
Schlosser & Blischak (2004)	ES	4 M	2 A 2 W	4 ASD	Direct Instruction	% of Correct Words—Speech % of Correct Words—Print % of Correct Words—Speech + Print	0.18 (0.25) 0.50 (0.28) 0.35 (0.27)	DNM
Schneider et al. (2013)	ES	4 M	NR	4 ASD	Strategy Instruction	TWW + SR TWW + HW Percentage CWS + SR Percentage CWS + HW Story Elements + SR Story Elements + HW	0.39 (0.26) 0.34 (0.34) 0.25 (0.32) −0.11 (0.22) 2.60 (0.43) 2.59 (0.38)	DNM
Stewart & Singh (1986)	HS	2 F 2 M	NR	3 ID 1 DD	Direct Instruction	# of Words Spelled Correctly	1.73 (0.25)	MWR
Stowitschek et al. (1987) Study 1	ES	NR	NR	4 ID	Self-Monitoring	# of Correct Letters	0.18 (0.14)	DNM

(continued)

Table 1. (continued)

	Participant characteristics				Intervention characteristics			
Study	Grade	Sex	Race	Diagnoses	IV	DV	SMD (*SE*)	Quality
Stowitschek et al. (1987) Study 2	ES	NR	NR	4 ID	Self-Monitoring	# of Correct Letters	0.09 (0.16)	DNM
Streitwieser (1982)	MS	14 F 13 M	NR	27 ID	Direct Instruction	# of Grammatically Correct Sentences	0.14 (0.37)	DNM
Woods-Groves et al. (2018)	PS	7 F 13 M	15 W 1 A	6 ASD 6 ID 5 DD 3 Other	Strategy Instruction	ANSWER Strategy Rubric	1.23 (0.47)	MWR
Woods-Groves et al. (2017)	PS	6 F 9 M	15 W	4 ASD 6 ID 1 DD 4 Other	Strategy Instruction	EDIT Total	0.78 (0.51)	MWR
Woods-Groves et al. (2015)	PS	5 F 14 M	19 W	9 ASD 9 ID 1 DD	Strategy Instruction	EDIT Total	1.29 (0.49)	M
Woods-Groves et al. (2014)	PS	5 F 14 M	19 W	10 ASD 9 ID	Strategy Instruction	ANSWER Strategy Rubric Analytic Rubric	8.20 (1.40) 0.91 (0.46)	MWR
Woods-Groves et al. (2012a)	PS	5 F 11 M	15 W 1 H	7 ASD 6 ID 1 DD 2 Other	Strategy Instruction	ANSWER Strategy Rubric	2.49 (0.66)	MWR
Woods-Groves et al. (2012b)	PS	8 F 8 M	14 W 2 B	5 ASD 7 ID 1 DD 3 Other	Strategy Instruction	ANSWER Strategy Rubric	1.79 (0.57)	M
Woods-Groves et al. (2020)	PS	7 F 9 M	15 W 1 A	4 ASD, 10 ID 1 DD, 1 Other	Strategy Instruction	Rubric Total	1.31 (0.53)	M

Note. IV = independent variable; DV = dependent variable; SMD = standardized mean difference; HS = high school; PS = postsecondary; MS = middle school; ES = elementary school; W = White; ASD = autism spectrum disorder (no intellectual disability); ID = intellectual disability; CWS = correct writing sequences; IWS = incorrect writing sequences; IEP = Individualized Education Program; SSS = spelling sensitivity score; SR = speech recognition; HW = handwriting; M = Meets without Reservations; B = Black; MWR = Meets with Reservations; A = Asian; DD = developmental disability other than ASD or ID (e.g., Down syndrome); NR = not reported; SRSD = self-regulated strategy development; TWW = total words written; DNM = Does Not Meet; H = Hispanic; SRD = systematic reinforcer delivery.

outcome, and the level of language at which the participants were writing. In addition, we conducted a meta-regression to examine the influence of moderating variables on the summary effect. We used the categorical WWC quality rating and average age of participants as moderators. We also included the dichotomous SRSD status of the intervention, the number of intervention sessions (averaged across participants in SCD studies), and the percentage of students in each study with ID. Given the unique characteristics of students with ID (e.g., working memory deficits and difficulty generalizing knowledge; Gargiulo & Bouck, 2018), we believed it important to account for effects of participants' intellectual functioning. All continuous moderators were evaluated for multicollinearity prior to conducting the meta-regression. We identified no problems.

Results

Participants and Settings

Four hundred and twenty-four participants (133 females, 277 males) were described in the 52 studies. Sex was not reported for 14 individuals across two studies. The average age of participants was 14.1 years old ($SD = 4.4$). Race and ethnicity demographics were reported for only 247 participants: White ($n = 207$), Black ($n = 22$), Asian ($n = 11$), and Hispanic ($n = 5$), Multiracial ($n = 2$). Fifteen studies included elementary school students, 14 included middle school students, 11 included high school students, and 10 included postsecondary students. Two studies only reported participants' ages. Most participants had ID ($n = 239$), followed by ASD ($n = 130$), ASD and ID (17), or another developmental disability (e.g., Down syndrome; $n = 17$). Only 24 studies reported IQ values for all participants. Of these, the average IQ was 62.9 ($SD = 17.8$). Twenty-one other participants in six group studies were reported as having a disability other than IDD and could not be disaggregated from the analysis. These individuals comprised between 6.3% and 38.1% of the studies' samples. Most studies ($n = 30$) took place in a public school, the majority of which were self-contained special education classrooms. Ten studies took place at a university, and four studies took place in a specialized facility for students with disabilities. In eight studies, participants received intervention in more than one location.

Study Characteristics

Most SCD studies used either multiple baseline (n = 18; 45.0%) or probe (n = 18; 45.0%) across participants designs. Other SCD studies used a reversal (n = 2; 5.0%) or alternating treatment design (n = 2; 5.0%). All group studies used a pretest/posttest design with intervention and control groups with random allocation to groups. The number of intervention sessions ranged from 3 to 60 (M = 13.6, SD = 10.1). Average instructional time could not be computed for 15 studies. From the available data, average instructional length was 496.3 min.

Study Quality

Studies were coded with the following WWC quality indicators: Does Not Meet Standards (n = 17), Meets with Reservations (n = 20), Meets without Reservation (n = 15). SCD studies that did not meet standards most often did not report inter-observer agreement for each measure, did sample observations from all study phases, or the reported agreement was insufficient (e.g., >.80). Group studies that did not meet standards often did not correct baseline-phase inequalities between intervention and control groups. Most studies that met with reservations were SCD studies that included only three data points for baseline.

Writing Intervention Effects

The summary effect for the 96 effect sizes was 1.44 (SE = 0.15, PI = −1.21 to 4.09), which was statistically significant ($p < .001$). The model identified statistically significant heterogeneity (Q = 1,730.31, df = 95, $p < .0001$, I^2 = 97.88%).

The results of the first sensitivity analysis revealed a substantial difference in the average effects for SCD (SMD = 1.47, SE = 0.19, PI = -0.69 to 3.63, $p < .001$) and group design studies (SMD = 1.11, SE = 0.37, PI = −2.65 to 4.87, p = .0117). The second sensitivity analysis showed that the 27 effects from the 10 SCD studies that did not collect intervention phase data were considerably higher than the summary effect from the SCD studies that did report such data (SMD = 2.45 vs. 0.98, respectively). In addition, the SCD studies that did not report intervention phase data yielded a larger prediction interval (PI = 0.01–4.90 vs. −0.51 to 2.46, respectively). These findings indicate that the overall summary effect should be interpreted with caution. We therefore included whether the study was an SCD study and whether the SCD study reported intervention phase data as control variables in the meta-regression.

Subgroup Analyses

There was considerable variation in effect sizes for independent and dependent variables and in Levels of Language (see Table 2). Strategy instruction was the most common intervention (22 studies; 52 effect sizes), followed by direct instruction (13 studies; 23 effect sizes). The remaining six intervention categories (i.e., response prompting, writing process, pre-writing, modeling alone, self-monitoring, and systematic reinforcer delivery interventions) were used infrequently and only contributed 21 effect sizes. All independent variable summary effects were significant (SMDs = 0.60 –2.48), except self-monitoring. Modeling alone (SMD = 2.48) and strategy instruction (SMD = 2.09) interventions yielded the largest effects. Writing process interventions yielded the smallest effect (SMD = 0.60). However, modeling alone studies only contributed three effect sizes, so this summary effect should be interpreted with caution.

We identified the following dependent measures: output (k = 21), quality (k = 18), conventions (k = 18), elements (k = 15), spelling (k = 11), handwriting (k = 6), procedures (k = 6), and organization (k = 1). All categories yielded significant summary effects (SMDs = 0.67–4.90), except for handwriting. Of these, organization and quality outcomes resulted in the largest effects (SMD = 4.90 and 2.29,

Table 2. Subgroup Effects by Independent Variable, Dependent Measure, and Level of Language.

Intervention characteristic	SMD	*SE*	PI
Independent variable			
Strategy instruction (k = 52)	2.09***	0.25	−1.22 to 5.41
Direct instruction (k = 23)	0.62***	0.11	−0.31 to 1.54
Response prompting (k = 8)	0.75***	0.17	−0.03 to 1.53
Writing process (k = 4)	0.60**	0.22	−0.13 to 1.33
Pre-writing (k = 3)	1.55*	0.76	−1.09 to 4.18
Modeling alone (k = 3)	2.48*	1.26	−1.94 to 6.91
Self-monitoring (k = 2)	0.14	0.10	−0.07 to 0.34
SRD (k = 1)	0.94*	0.41	0.14 to 1.75
Dependent measure			
Output (k = 21)	1.14***	0.20	−0.54 to 2.83
Quality (k = 18)	2.29***	0.44	−1.09 to 5.68
Conventions (k = 18)	0.67***	0.12	−0.23 to 1.58
Elements (k = 15)	2.27***	0.37	−0.36 to 4.89
Spelling (k = 11)	0.69***	0.20	−0.57 to 1.95
Handwriting (k = 6)	0.93	0.59	−1.84 to 3.70
Procedures (k = 6)	2.46*	1.19	−3.69 to 8.61
Organization (k = 1)	4.90**	1.55	1.85 to 7.94
Level of language			
Letter (k = 2)	0.13	0.10	−0.07 to 0.34
Word (k = 15)	0.76***	0.20	−0.59 to 2.11
Sentence (k = 12)	0.50**	0.17	−0.59 to 1.58
Discourse (k = 67)	1.82***	0.20	−1.13 to 4.77
Persuasive (k = 24)	2.06***	0.25	−0.12 to 4.24
Narrative (k = 22)	1.31***	0.27	−1.03 to 3.64
Informative (k = 17)	2.33***	0.58	−2.27 to 6.92
Other (k = 4)	1.13***	0.16	−0.55 to 1.70

Note. SMD = standardized mean difference; PI = prediction interval; k = number of effect sizes; SRD = systematic reinforcer delivery.
*$p < .05$. **$p < .01$. ***$p < .001$.

respectively). However, there was only one effect size for organization, so this should be interpreted with caution. Conventions and spelling outcomes yielded the smallest effects (SMD = 0.67 and 0.69, respectively).

All levels of language were represented in the studies: letter ($k = 2$), word ($k = 15$), sentence ($k = 12$), and discourse ($k = 67$). The summary effects for all levels were significant, except letter (SMDs = 0.13–1.82). Discourse yielded the largest effect (SMD = 1.82). The 67 discourse-level effect sizes stemmed from persuasive ($k = 24$), narrative ($k = 22$), informative ($k = 17$), and other writing tasks ($k = 4$). All effects were significant (SMDs = 1.13–2.33).

Meta-Regression

When controlling for study type and whether the study collected intervention-phase data, the omnibus test of the coefficients was significant ($Q = 39.24$, $df = 8$, $p < .0001$) indicating that the moderators led to meaningful changes in the summary effect. SRSD studies were significantly associated with larger effect sizes (SMD = 1.66, $SE = 0.41$, $p = .00467$), as were studies that met the WWC standards with and without reservations when compared with studies that did not meet standards (SMD = 0.85, $SE = 0.35$, $p = .02986$ and 1.14, $SE = 0.36$, $p = .00726$, respectively). Overall, the moderators accounted for 27.33% of the variance. However, the test for residual heterogeneity was statistically significant ($Q = 1039.80$ $df = 77$, $I^2 = 96.76\%$, $p < .0001$) indicating that considerable heterogeneity remained unexplained.

Discussion

In this study, we used meta-analysis to address the following research questions:

> **Research Question 1**: What are the writing interventions that have been used with students with IDD and how were writing skills measured?
> **Research Question 2**: What are the effects of those interventions?
> **Research Question 3**: How do study characteristics moderate those effects? We discuss the answers to these questions below within the context of prior writing intervention research.

What Interventions Have Been Used and What Writing Skills Were Measured?

We identified a diverse body of writing literature for students with IDD regarding design, independent and dependent variables, and study quality. As in previous work (Gillespie & Graham, 2014), we found that strategy instruction was the most frequent intervention and yielded the largest effects. More than half of the included studies used the SRSD framework. Relatedly, most interventions focused on discourse-level writing skills; very few included studies targeted writing skills at the word or sentence levels. This is interesting, given the sentence level writing challenges documented for students with IDD (Rodgers & Datchuk, 2020).

The included studies reported 47 unique writing outcomes, of which output, conventions, and quality were the most common. Prior writing reviews (Kaldenberg et al., 2016) identified 39 unique outcome measures across 23 studies, the most common of which were the number of functional elements, holistic rubrics, and total words (which correspond to elements, output, and quality in the current study). These findings demonstrate that writing research on students with IDD, as in writing research more broadly, targets a wide range of important writing skills.

Finally, the quality of the writing research for students with IDD is mixed. Included studies were almost evenly spread across the quality categories, though the fewest number of included studies were categorized as Met Without Reservations. We identified several common quality-related issues, including not reporting interobserver agreement, insufficient agreement, and inequality of intervention and control groups. These issues are not unique to this population; Graham et al. (2016) reported that less than half of the included studies had acceptable reliability rates. Interobserver agreement and baseline equality are critical aspects of sound methodology, so this missing information is troubling. We are less confident in the results of these studies, and the results of this study should be interpreted with those limitations in mind.

What Are the Effects of Writing Interventions Used With Students With IDD?

The writing interventions were generally effective at improving writing outcomes. The overall summary effect was 1.44 ($SE = 0.15$), suggesting that, on average, participants improved writing outcomes by over one standard deviation. The prediction interval (SMD = −1.21 to 4.09) indicates that the effects are widely distributed. Descriptively, our results align with previous narrative reviews for students with IDD (e.g., Joseph & Konrad, 2009; Rogers & Graham, 2008). Quantitatively, our results are also similar to meta-analyses of writing interventions for individuals with other disabilities. For example, prior meta-analyses for students with learning disabilities have yielded effect sizes as high as 0.74 and 1.15 (Gillespie & Graham, 2014; Graham & Perin, 2007). Our results are promising, suggesting that students with IDD benefit from targeted writing instruction in ways comparable with students with less significant needs.

SCD studies yielded a larger summary effect than group studies (SMD = 1.47 vs. 1.11), but these results should be interpreted with caution. SCD studies may have described fundamentally more effective interventions than those in group studies, but it is also possible that the individual and small group instruction inherent in SCD studies and/or design-specific phenomena (e.g., SCD inherently yields larger and possibly biased effects) might also be responsible. We also found that SCD studies that omitted intervention phase data yielded considerably larger effect sizes than SCD studies that did report such data (SMD = 2.45 vs. 0.98). This was a source of bias. Prior meta-analyses did not include SCD studies, so our findings cannot be easily situated in prior work and are unique in this regard.

Regarding intervention types, modeling and strategy instruction yielded the largest summary effects. However, because there were only three modeling alone effect sizes, we are more confident in the estimate for the strategy instruction interventions, (k = 52). Like prior work (e.g., Asaro-Saddler, 2016; Gillespie & Graham, 2014), we found that SRSD studies were associated with larger effect sizes than non-SRSD studies. Importantly though, the intervention categories with the smallest effect sizes (writing process and direct instruction) were still moderate (SMD = 0.60 and 0.62, respectively). In previous research, Datchuk and colleagues (2020) documented that direct instruction and SRSD studies led to equal gains in correct writing sequences (CWS) but that direct instruction did so more efficiently. Pennington and Delano (2012) also reported the positive effects of direct instruction.

How Do Study Characteristics Moderate the Effects?

Several factors moderated the effects of the interventions. In addition to the SRSD effects described above, we found that higher study quality was associated with larger effects. Specifically, studies that were classified as Met with Reservations were associated with effect sizes 0.85 standard deviations larger than those studies that did not meet the standards, and studies that Met without Reservations were associated with effect sizes that were 1.14 standard deviations larger than those that did not meet standards. Our results diverge from prior research in this respect. Prior studies reported that study quality was not significantly correlated with effect size magnitude, nor did study quality significantly moderate the summary effect (Gillespie & Graham, 2014; Graham & Perin, 2007). However, the previous studies presented quality as a percentage, whereas we categorized studies into one of the three WWC categories. Differences in our results may be at least partially attributable to these differing approaches. Average age, number of intervention sessions, and percentage of students with ID were not statistically significant, which suggests that students may have benefited from intervention equally, regardless of age or disability type. Students with ID benefited from writing interventions in a comparable way to students with other developmental disabilities without comorbid ID.

Limitations

The findings described in this article should be interpreted within the context of several limitations. First, the SCD studies that did not collect intervention phase data yielded larger effect sizes than studies that did collect such data. These effect sizes likely contain bias. Second, we excluded SCD studies that were incongruous with the calculation of the BC-SMD effect size. This included studies at the letter and sentence levels of language. Such exclusion also introduces bias but was necessary for the calculation of the BC-SMD, which was essential to the analyses conducted in this article. Finally, publication bias may have impacted the results, especially because we did not include theses.

Future Directions

The results of this meta-analysis provide several avenues for future research. First, regarding study design, future researchers should include intervention phase data when reporting their work. Studies that do not are likely biased, which impacts both the primary study and secondary data analysis (such as in this meta-analysis). Second, future researchers may wish to describe facets of intervention intensity more thoroughly—especially for SCD studies. We attempted to calculate average intervention time per study to use as a moderator in our meta-regression, but too few studies reported the necessary information. Third, future researchers may wish to develop more interventions at the word and sentence levels of language. The majority of studies that we identified were at the discourse level. However, this paucity is partly an artifact of our analytic plan; some of these studies did not meet the requirements for the BC-SMD, which was critical to this article. Future researchers could descriptively aggregate such studies to explore the important contributions of such work.

Implications for Practice

Prioritizing the development of writing skills in students with IDD can help them meet their academic, employment, independent living, and social goals. Fortunately, the results of this meta-analysis suggest that these learners can and do benefit from writing instruction. Strategy instruction (and SRSD studies especially), response prompting, and direct instruction interventions all led to significant improvements in writing outcomes. In addition, students with more moderate needs, such as those with ID, benefited in a comparable way

to those with less moderate needs, such as students with ASD without comorbid ID. Thus, these instructional approaches have a great deal of promise, and educators should consider them when working on writing with students with IDD to improve both current and long-term outcomes.

Authors' Note

Derek B. Rodgers is now affiliated with the College of Education, Department of Teaching and Learning, University of Iowa.

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ORCID iD

Susan J. Loveall https://orcid.org/0000-0002-3851-8433

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*Indicates that the paper was included in the meta-analysis.

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